

The Hypogamy Puzzle

Why the educational reversal did not (yet) flip the marriage market

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Motivation

- **Hypergamy** — women “marrying up” — has long defined marriage markets.
 - Strong focus on **same-type** couples (positive assortative matching, **PAM**) in research.
 - Yet women now *outpace* men in higher education across the developed world.
- Mechanically, traditional hypergamy and PAM is no longer feasible for many highly-educated women.

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 - Yet women now *outpace* men in higher education across the developed world.
- Mechanically, traditional hypergamy and PAM is no longer feasible for many highly-educated women.
- **The puzzle: hypogamy** — women “marrying down” — is rising, but far less than the educational reversal would predict.

Question

What solves the hypogamy puzzle — composition, the labor market, home production, or asymmetric preferences over who marries whom?

Why is this important?

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Three contributions

1. Document the rise of hypogamy in Danish registers, 1980–2013; adjust for compositional change via a likelihood-ratio measure.
2. Show that hypogamous couples face higher but *stable* divorce risk, while hypergamous and assortative unions exhibit *rising* dissolution rates.
3. Build a structural search-and-matching model to study the determinants and consequences of hypogamy—quantifying the roles of composition, preferences, household technology, and the labor market.

- **End of hypergamy / trends:** Esteve et al. (2016); Almås et al. (2020); Bertrand–Kamenica–Pan (2015).
- **Measuring sorting:** Eika–Mogstad–Zafar (2019); Chiappori–Costa Dias–Meghir–Zhang (2025); Almar–Schulz (2024); Almar–Friedrich–Reynoso–Schulz–Vejlin (2025b).
- **Returns to sorting:** Chiappori–Salanié–Weiss (2017); Almar–Friedrich–Reynoso–Schulz–Vejlin (2025a).
- **Frictional matching:** Becker (1973); Shimer–Smith (2000); Goussé–Jacquemet–Robin (2017, 2025); Holzner–Schulz (2023); Foerster–Obermeier–Schulz (2025).

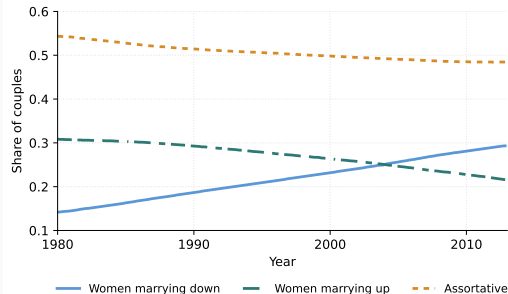
Our contribution: a frictional marriage-market model with *asymmetric* type-dependent love-shock locations, calibrated to Danish administrative data spanning 30+ years.

Danish administrative registers, Statistics Denmark, 1980–2024.

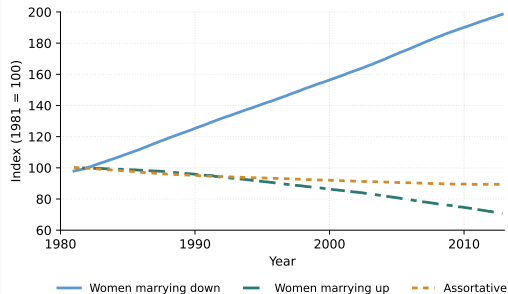
- **Sample:** ages 19–60, cohorts born ≤ 1994 ; analysis frozen at 2013 to keep composition stable.
- **Education** (UDDA): four ISCED-based levels — primary, secondary, bachelor, master/PhD; assigned at age 30.
- **Partnership** (BEF): legal marriage *and* co-residential cohabitation; opposite-sex; age gap < 15 years.
- Wages and individual labour supply from linked registers.

Fact 1 — The rise of hypogamy

Couple-type shares (levels)



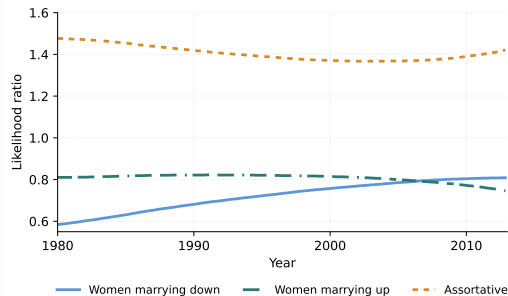
Indexed (1981 = 100)



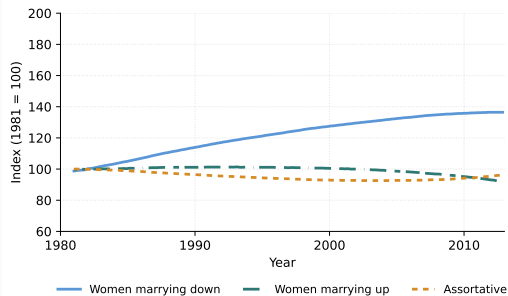
- Hypogamy doubles: 15% (1980) → 30% (2013); hypergamy falls by ~one-third; assortative matching slightly down.

Fact 2 — Adjusting for compositional change

Likelihood ratio (levels)



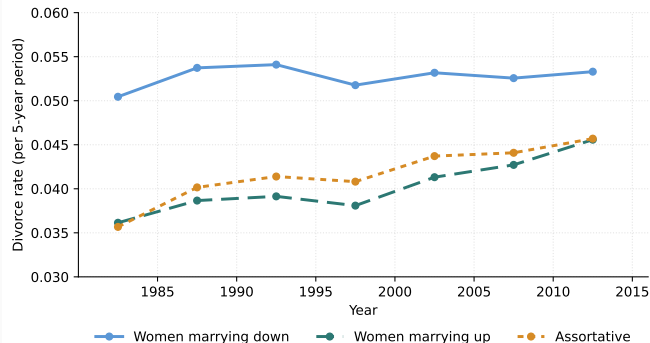
Indexed (1981 = 100)



$s(i, j) = \frac{P(e_m=i, e_f=j)}{P(e_m=i) P(e_f=j)}$ purges composition: hypogamy still rises $\approx 40\%$; hypergamy and assortativeness flat $\Rightarrow$ **behavioural change**, not just supply.

► decomposition by woman's education

Fact 3 — Divorce rate by couple type



Divorce rate, by type

For couple type

$T \in \{\text{hypo}, \text{hyper}, \text{assort}\}$:

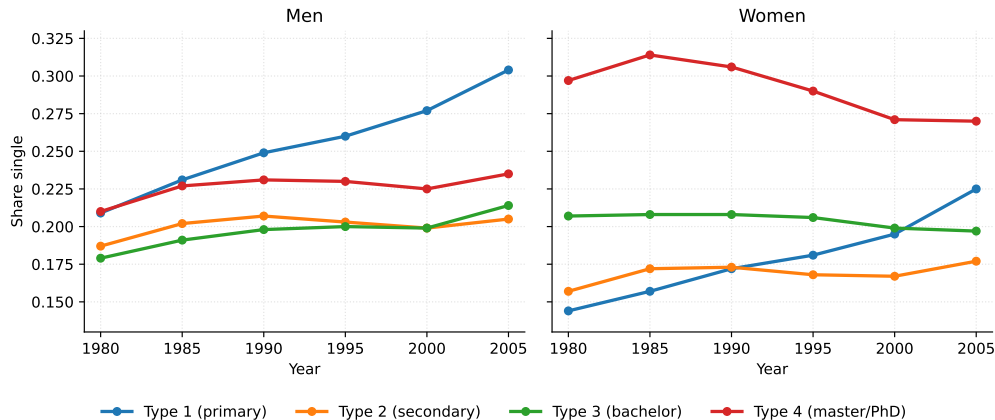
$$\text{DR}_T = \frac{\sum_{(i,j) \in T} \text{outflow}(i,j)}{\sum_{(i,j) \in T} \mu(i,j)},$$

probability a couple of type T dissolves over the 5-year period.

- Hypogamy *high and stable* at $\sim 5\text{--}5.4\%$.
- Hypergamy and assortative *rise from below* ($3.6\% \rightarrow 4.6\%$) and converge.

► marriage rate

Fact 4 — Singles by type



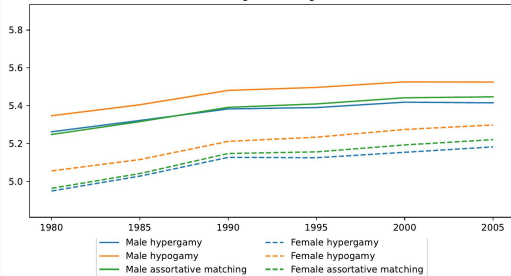
Supply-side perspective of the hypogamy puzzle. Hypogamy *rises* (Fact 1) but doesn't go far enough: Type-4 women increasingly find partners (single share 30% → 27%), while Type-1 men are largely bypassed and stay single in growing numbers (21% → 30%).

Fact 5 — Wages and labour supply by sorting pattern

Average log residual wages

Average log residual wages by sorting pattern

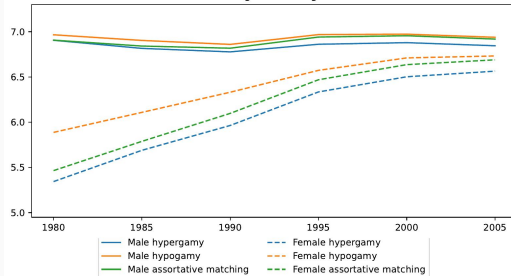
Weighted average



Average labour hours

Average labor hours by sorting pattern

Weighted average



- Hypogamous couples have the highest wages — for *both* partners.
- Female labour hours rise sharply 1980–2005; hypogamous women remain on top.
- → wages *exogenous*, labour supply *endogenous* in the model.

Model — setup (1/4)

We need a model that delivers: (i) rising hypogamy beyond composition; (ii) *asymmetric* hypo-vs-hyper response; (iii) divergent divorce-rate dynamics across couple types.

Continuous-time search–matching market à la Goussé–Jacquemet–Robin (2017, 2025); Holzner–Schulz (2023); Shimer–Smith (2000).

- Unit measures of men and women; types $i, j \in \{1, 2, 3, 4\}$, observed and constant.
- Each five-year period (1980–85, . . . , 2005–10) is treated as a *separate* steady state.

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- Each five-year period (1980–85, . . . , 2005–10) is treated as a *separate* steady state.
- Cobb–Douglas meeting technology:

$$\lambda(N_m, N_f) = \xi (N_m N_f)^{-1/2}.$$

- Marriages dissolve at rate δ when an updated love shock z flips the surplus negative.
- ξ, δ are *externally* pinned by stock-weighted least squares on flows.

Model — home production (2/4)

CES home production

$$q_1(d_m, d_f) = z \cdot [K_m d_m^\rho + K_f d_f^\rho]^{1/\rho}, \quad \rho < 1, K_m + K_f = 1.$$

$\sigma = 1/(1 - \rho)$: elasticity of substitution between male and female domestic time.

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Household problem. Choose (d_m, d_f) to maximise $q_1 \cdot X_1$ subject to

$$X_1 = w_m(1 - d_m) + w_f(1 - d_f) - A_m(i) - A_f(j) - C,$$

the consumption residual after market work, fixed costs A , and marriage cost C .

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Deterministic maximised output (love shock z factored out):

$$c_{xy}(i, j) \equiv \frac{q_1 X_1}{z} = \frac{Y(i, j)^2}{4 P_H(i, j)} \implies q_1 X_1 = z \cdot c_{xy}(i, j).$$

The shock z scales the maximised product multiplicatively; $z \cdot c_{xy}$ enters the surplus.

$Y(i, j) = w_m(i) + w_f(j) - A_m(i) - A_f(j) - C$ disposable income net of fixed and marriage costs;

$P_H(i, j) = [K_m^\sigma w_m(i)^{1-\sigma} + K_f^\sigma w_f(j)^{1-\sigma}]^{1/(1-\sigma)}$: CES unit cost of home output.

Model — love shock (3/4)

Pareto love shock — shape θ , type-dependent scale $z_m(i, j)$:

$$z \sim \text{Pareto}(\theta, z_m(i, j)), \quad \Pr(z > \bar{z} \mid i, j) = (z_m(i, j)/\bar{z})^\theta, \quad \bar{z} \geq z_m.$$

Asymmetric type-dependent scale

$$\log z_m(i, j) = \zeta_0 + \zeta_{\text{diag}} \min(i, j) + \zeta_{m\text{-hyp}} \max(i - j, 0) + \zeta_{f\text{-hyp}} \max(j - i, 0).$$

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Interpretation

- ζ_{diag} — supermodular core; drives positive assortative matching on the diagonal.
- $\zeta_{m\text{-hyp}}$ — per-step penalty / premium on male-hypergamous cells ($i > j$).
- $\zeta_{f\text{-hyp}}$ — per-step penalty / premium on female-hypergamous cells ($j > i$).
- The two off-diagonal slopes are *signed and independent* $\Rightarrow$ accommodates asymmetric LR(1,4) vs. LR(4,1).

Marital surplus (option-value form)

$$(r + \delta) S(i, j; z) = z c_{xy}(i, j) - rV_m^0(i) - rV_f^0(j) + \delta \bar{S}(i, j).$$

Acceptance (with reservation shock $\bar{z}$): $\alpha(i, j) = (z_m(i, j)/\bar{z}(i, j))^\theta$ when $\bar{z} > z_m$.

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Steady-state marriage distribution

$$\mu(i, j) = \frac{\lambda}{\delta} \cdot \frac{\alpha(i, j)}{1 - \alpha(i, j)} n_m(i) n_f(j).$$

Estimation. SMM on **81 moments** (aggregate marriage rate, married marginals, 16 LR cells, 16 cell-wise divorce rates, labour supply by cell). **22 internal parameters:**

($\zeta_0, \zeta_{\text{diag}}, \zeta_{m\text{-hyp}}, \zeta_{f\text{-hyp}}, \theta, \rho, K_m, K_{s,m}, K_{s,f}$, quadratic-in-wage A polynomials, marriage cost C).

Calibration — likelihood ratios (1980–85)

Empirical LR				
$e_m \backslash e_f$	1	2	3	4
1	1.50	0.60	0.38	0.12
2	0.86	1.29	0.81	0.41
3	0.38	1.05	3.08	1.49
4	0.22	0.84	3.00	10.97

Model LR				
$e_m \backslash e_f$	1	2	3	4
1	1.60	0.52	0.34	0.18
2	0.64	1.45	0.91	0.49
3	0.40	0.88	3.72	1.91
4	0.36	0.76	3.26	12.45

- Diagonal (**bold**) sorting strength captured: $(4, 4)$: $11.0 \rightarrow 12.4$; $(3, 3)$: $3.1 \rightarrow 3.7$.
- Asymmetry of off-diagonal pattern survives.
- Extreme corners $(1, 4), (4, 1)$ overstated by $\sim 50\%$.

- Hypogamy rises in Denmark even after purging composition — *behaviour*, not just supply.
- Divorce dynamics diverge across couple types: hypogamy stable, others rising.
- Structural model with *asymmetric* love-shock locations and CES home production rationalises both the LR matrix and divorce dispersion.
- **Next steps:** estimate on all time windows, counterfactuals and decompositions, robustness on income-/SES-based types, welfare implications, intra-household allocations.

Wages here are exogenous. In the hypogamy paper (and GJR), $w_m(i)$, $w_f(j)$ are type-specific endowments — they “fall from the sky.”

A natural extension: McCall-style job search.

- Gender- and type-specific wage offer distributions $F_g(w \mid \text{type})$.
- Couple-type-specific reservation wages $\bar{w}(i, j)$.
- Labor and marriage transitions jointly determined.

Example: Holzner & Schulz (2023). Simultaneous search in labor and marriage markets; documents “labor-market divorces” — married women’s transitions into employment can reduce marital surplus and raise divorce risk.

Still missing in this literature: FIRMS.

Bringing firms in.

- **Almar–Friedrich–Reynoso–Schulz–Vejlin (2025b)**. Firms make training and promotion decisions that interact with families' marriage formation and labor supply decisions. *No labor-market search frictions*.
- **Foerster–Obermeier–Schulz (2025)**. Establishment closures as a quasi-experimental shock — displacement can dissolve marriages and reshape remarriage types. Firms affect the marriage market through the displacement channel.
- **Denderski–Kaas–Schulz–Siassi (2026)**. Couples' joint job search with firms posting vacancies *and* labor-market frictions; quantifies welfare effects of joint vs. separate progressive taxation and a suite of UI reforms. However, no marriage market.

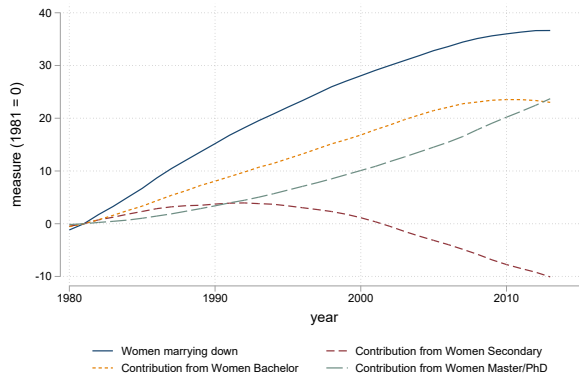
Next step.

- Integrate *heterogeneous firms* with on-the-job search.
- Study how firms' hiring and promotion / retention strategies interact with family decisions.
- Quantify how this shapes inequality *across genders and across households*.

Thank you.

Comments and questions welcome.

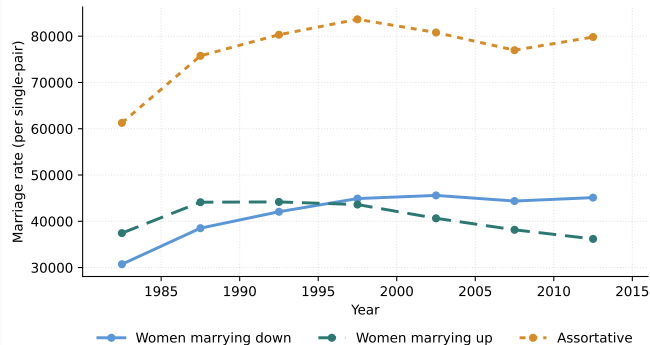
Appendix — hypogamy decomposition by woman's education



Decomposition by woman's education

- Bachelor and Master/PhD women are the engine of rising hypogamy.
- Contribution of *secondary*-educated women is *negative* from ~ 1995 .
- Consistent with the educational reversal: more highly-educated women $\Rightarrow$ more hypogamous matches.

Appendix — marriage rate by couple type



Marriage rate, by type

For couple type T :

$$\text{MR}_T = \frac{\sum_{(i,j) \in T} \text{inflow}(i,j)}{\sum_{(i,j) \in T} n_m(i) n_f(j)},$$

n_m, n_f : within-gender shares of singles (population at risk).

- Assortative MR $\gg$ off-diagonal $\Rightarrow$ strong positive sorting at *formation*.
- Hypogamy rises steadily; hypergamy declines after 1990.
- **Crossing ~ 1995** : hypogamy becomes the dominant off-diagonal flow.

Appendix — calibration: married labour hours (1980–85)

Male hours h_m^1 , empirical				
$e_m \backslash e_f$	1	2	3	4
1	6.76	6.93	6.87	6.67
2	6.88	7.01	6.90	6.49
3	7.16	7.20	7.19	7.05
4	7.11	7.10	7.06	7.03

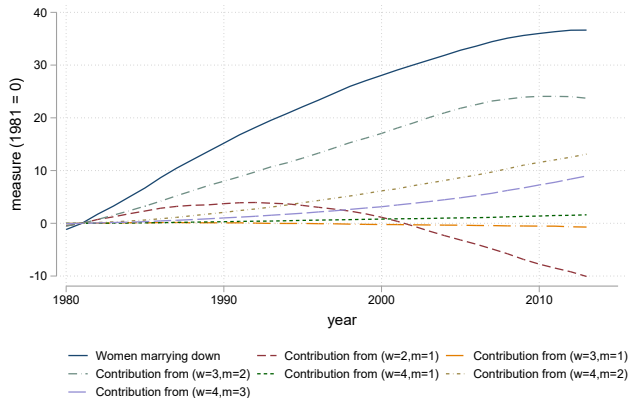
Female hours h_f^1 , empirical				
$e_m \backslash e_f$	1	2	3	4
1	5.20	5.65	6.00	5.84
2	5.32	5.56	6.16	6.25
3	5.23	5.35	6.42	6.23
4	5.05	5.22	5.81	6.27

Male hours h_m^1 , model				
$e_m \backslash e_f$	1	2	3	4
1	6.47	6.54	6.62	6.80
2	6.99	7.04	7.12	7.23
3	7.26	7.30	7.27	7.41
4	6.87	6.87	6.97	6.99

Female hours h_f^1 , model				
$e_m \backslash e_f$	1	2	3	4
1	5.20	5.27	5.65	6.18
2	5.64	5.70	6.08	6.67
3	5.62	5.69	6.16	6.72
4	4.81	4.93	5.26	6.00

Most cells within ± 5 – 10% ; gradient in e_f (women's hours rising in own-type) captured; bottom row ($e_m=4$) understated.

Appendix — cell-level hypogamy decomposition



Contribution by (e_f, e_m) cell

- Largest pulls upward:
 $(w = 3, m = 2)$ and
 $(w = 4, m = 2)$ — bachelor-
and master-educated women
paired with
secondary-educated men.
- Negative drag from
 $(w = 2, m = 1)$ from ~ 2000 .
- Confirms decomp. by
woman's education:
high-type women with
secondary-educated partners
drive the rise.

Appendix — moment vector (81 entries)

- Aggregate marriage rate (1)
- Married marginals $m_m(i)$, $m_f(j)$ (4+4)
- Cell-wise likelihood ratios $LR(i,j)$ (16)
- Cell-wise divorce rates $\delta(i,j)$ (16)
- Matched labour supply $h_m^1(i,j)$, $h_f^1(i,j)$ (16+16)
- Singles' labour supply $h_m^0(i)$, $h_f^0(j)$ (4+4)

Appendix — external calibration of ξ, δ

Stock-weighted least-squares regression on flows (Goussé moment):

$$1 = \beta_1 \cdot \text{MR}(i, j) \sqrt{N_m N_f} + \beta_2 \cdot \text{DR}(i, j) + \varepsilon_{ij},$$

with weights $w_{ij} = n_m(i) n_f(j)$. Recovers $\hat{\xi} = 1/\hat{\beta}_1$, $\hat{\delta} = 1/\hat{\beta}_2$. WLS preferred over OLS because high-mass cells dominate; honest SEs and higher R^2 .

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